

Bonding

1. What is chemical bonding? Explain ionic, covalent and metallic bonds.

Chemical bonding is the force or interaction that holds atoms together in a compound. Atoms form bonds to achieve a more stable electron arrangement, usually a filled valence shell or an octet of valence electrons. They achieve this stability by sharing, gaining, or losing electrons.

There are three major types of chemical bonds:

- **Ionic bond:** An ionic bond is the electrostatic attraction between oppositely charged ions, a cation and an anion. It generally forms between a metal and a non-metal. The metal loses electron(s) and becomes a cation, while the non-metal gains electron(s) and becomes an anion.
- **Covalent bond:** A covalent bond is formed when two atoms share electrons through overlapping orbitals. It generally occurs between two non-metals or between a metalloid and a non-metal.
- **Metallic bond:** A metallic bond consists of a mobile “sea” of delocalized electrons that is attracted to a cluster of metal cations. The delocalized electrons act as the glue holding the metal cations together.

2. How is a chemical bond formed? What is the role of electronegativity?

Atoms form chemical bonds to obtain a stable electron arrangement, usually a filled valence shell or an octet. They achieve this stability by gaining, losing, or sharing electrons. Electronegativity (EN) is the measure of the tendency of an atom to attract or “steal” an electron from another atom within a chemical bond. Metals generally have low electronegativity, whereas non-metals have high electronegativity.

The difference in electronegativity helps determine the type of bond:

- Nearly identical EN: Pure covalent bond.
- Moderate difference in EN: Polar covalent bond.
- Large difference in EN: Ionic bond.

Thus, electronegativity plays an important role in determining whether electrons are shared equally, shared unequally, or transferred between atoms.

3. What are single and double covalent bonds?

A single covalent bond is formed when two atoms share one pair of electrons.



A double covalent bond is formed when two atoms share two pairs of electrons.



The shared electrons in covalent bonding result from the overlap of orbitals between the bonded atoms.

4. What type of bonds are present in NaCl and methane?

NaCl: Sodium chloride contains an ionic bond. Sodium loses an electron to become Na^+ , while chlorine gains the electron to become Cl^- . The electrostatic attraction between Na^+ and Cl^- forms the

ionic bond. Methane (CH₄): Methane contains covalent bonds. Carbon and hydrogen share electrons to form C—H bonds. Since covalent bonds are formed through sharing of electrons, methane is a molecular compound.

5. If hydrogen bonding is present, how are boiling point and melting point affected?

Hydrogen bonding is an intermolecular attractive force. Stronger attractions between molecules require more energy to separate the molecules. Therefore, when hydrogen bonding is present and significant, the substance generally requires more energy for phase changes, resulting in higher boiling point and melting point compared with similar substances having weaker intermolecular attractions. Molecular compounds have relatively low melting points because attractions between separate molecules are weak; therefore, stronger intermolecular attractions require more energy to separate molecules.

6. What is the difference between hydrogen bonding in ethanol and methane?

Ethanol (C₂H₅OH) contains an O—H group. Oxygen is highly electronegative, so the O—H bond produces charge separation, allowing ethanol molecules to form hydrogen-bond attractions with one another. Methane (CH₄) does not contain an O—H, N—H, or similar strongly polar hydrogen-containing group. Therefore, methane does not form significant hydrogen bonds between its molecules. As a result, ethanol has stronger intermolecular attractions than methane and consequently requires more energy to separate its molecules.

7. What is a metallic bond?

A metallic bond is the attraction between metal cations and a mobile sea of delocalized electrons. The electrons are not associated with only one particular atom; instead, they are shared throughout the cluster of metal cations. These mobile electrons act as the “glue” holding the metal atoms together. Metallic bonding explains why metals are good conductors of electricity and heat, malleable and ductile, and lustrous.

Acids and Bases / Acidity

1. Write the definitions of Arrhenius acid, Arrhenius base, Brønsted–Lowry acid, Brønsted–Lowry base, Lewis acid, Lewis base and amphoteric substance.

Arrhenius Acid: An Arrhenius acid is a substance that contains hydrogen and releases hydrogen ions (H⁺) when dissolved in water.



Arrhenius Base: An Arrhenius base is a substance that releases hydroxide ions (OH[−]) when dissolved in water.



Brønsted–Lowry Acid: A substance that can donate a proton (H⁺) is called a Brønsted–Lowry acid.

Brønsted–Lowry Base: A substance that can accept a proton (H⁺) is called a Brønsted–Lowry base.

Lewis Acid: An electron-pair acceptor. Lewis Base: An electron-pair donor. Amphoteric substance:

A substance that can act as either an acid or a base is called an amphoteric substance. Water is an example.

2. What is a conjugate base?

A conjugate base is formed when an acid donates a proton (H^+).



Here, HA is the acid and A^- is its conjugate base. For example:



Here, CH_3COOH is the acid and CH_3COO^- is the conjugate base.

3. Define strong acid and weak acid with examples.

Strong acid: A strong acid is an acid that is completely ionized when dissolved in water. Examples include HCl, HNO_3 , HBr, HI and H_2SO_4 . Weak acid: A weak acid is an acid that is partially ionized in its solution. Examples include acetic acid (CH_3COOH), formic acid ($HCOOH$), HF and HCN.

4. What is the equilibrium constant? Explain the acid dissociation constant (K_a).

Equilibrium constant (K_c) expresses the relationship between the concentrations of products and reactants at equilibrium. For a general reaction:



$$K_c = [C]^c [D]^d / [A]^a [B]^b$$

For the ionization of a weak acid, the equilibrium constant is called the acid ionization constant (K_a). For acetic acid:



$$K_a = [H_3O^+][CH_3COO^-] / [CH_3COOH]$$

For acetic acid, $K_a = 1.75 \times 10^{-5}$.

Electrochemistry

1. What is electrochemistry?

Electrochemistry is the branch of chemistry that studies the relationship between electricity and chemical reactions, particularly the transfer of electrons. It includes the production of electricity from spontaneous chemical reactions and the use of electricity to bring about non-spontaneous chemical changes.

2. Explain the two types of electrodes with details and reactions.

Cathode: The cathode is the electrode where reduction occurs. At the cathode, electrons are accepted by the reacting species.



Here, Cu^{2+} gains electrons and is reduced to copper metal. Anode: The anode is the electrode where oxidation occurs. At the anode, electrons are generated because the substance loses electrons.



Here, zinc loses electrons and is oxidized to Zn^{2+} . Easy way to remember: AN OX — Anode = Oxidation; RED CAT — Reduction = Cathode.

3. What is an electrochemical cell?

An electrochemical cell is a system in which chemical and electrical energy are interconverted. It contains an anode and a cathode, because electrons produced at the anode must be consumed by the cathodic reaction. The electrodes are placed in an electrolyte through which ions can move.

There are two important types: (1) Galvanic/Voltaic cell: Generates electrical current from a spontaneous chemical reaction. (2) Electrolytic cell: Uses externally supplied electrical current to drive a non-spontaneous chemical reaction.

4. Explain anode and cathode reactions, oxidation and reduction reactions.

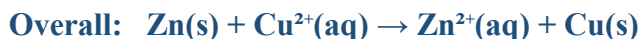
At the anode, oxidation occurs. The substance loses electrons.



At the cathode, reduction occurs. The reacting ion gains electrons.



These two reactions occur together because the electrons released during oxidation are consumed during reduction. The overall process is called a redox reaction. For a Daniell cell:



5. What is transport number? Write its mathematical expressions.

A transport number (transference number) represents the fraction of the total electric current carried by a particular ion in an electrolytic solution. It indicates how effectively an individual ion contributes to the total conductivity.

$$\text{For a cation: } t_{+} = I_{+} / (I_{+} + I_{-}) \text{ or } t_{+} = u_{+} / (u_{+} + u_{-})$$

$$\text{For an anion: } t_{-} = I_{-} / (I_{+} + I_{-}) \text{ or } t_{-} = u_{-} / (u_{+} + u_{-})$$

Where I_{+} = current carried by cation; I_{-} = current carried by anion; u_{+} = mobility of cation; and u_{-} = mobility of anion.

$$t_{+} + t_{-} = 1$$